CLINICAL · APPROACH TO

Approach to loss of smell

Loss of smell was a neglected symptom until COVID-19 made it famous. It matters more than patients and doctors once thought: it removes a warning system for smoke, gas and spoiled food, flattens the taste of food (most "taste" is smell reaching the nose from the mouth), affects mood and relationships, and in older people predicts both neurodegenerative disease and death. The approach divides causes into **conductive** loss, where odours cannot reach the olfactory cleft, and **sensorineural** loss, where the receptors or their central pathways are damaged, and looks for the treatable causes of each.

AT A GLANCE

- Conductive: polyps, CRS — oral steroid trial helps.
- Sensorineural: post-viral, head injury, age, Parkinson's.
- Psychophysical testing (Sniffin' Sticks, UPSIT).
- Olfactory training twice daily for months.

1 CHAPTER ONE Terms and anatomy

What can go wrong, and where.

1.1 Definitions

Anosmia is total loss of smell; **hyposmia** is a reduced sense of smell. **Parosmia** is a distorted perception of a familiar odour, usually unpleasant (coffee smells of sewage); **phantosmia** is perceiving a smell that is not there. Parosmia is often a sign that the olfactory nerve is regenerating and carries a reasonable prognosis.

1.2 How smell works

The olfactory epithelium lies in the narrow olfactory clefts, on the upper septum, the superior turbinate and the cribriform plate. About 10 to 15% of inspired air reaches it with each breath, more with sniffing. The receptor neurons send axons through the cribriform plate to the olfactory bulb and regenerate throughout life

from basal stem cells, which is why recovery is possible. **Retronasal** smell, from food in the mouth reaching the nose, makes up most of what patients call flavour, so a patient with anosmia often complains of losing taste.

DO NOT MISS

- **Unilateral loss with nasal blockage or bleeding:** a sinonasal tumour, including olfactory neuroblastoma.
- **Loss with headache, personality change or visual symptoms:** a frontal or olfactory groove meningioma.
- **Phantosmia in brief episodes:** temporal lobe seizures.
- **Gradual loss in an older adult with tremor or memory problems:** Parkinson's disease or dementia, in which smell loss can come years first.
- **Absent smell from birth with delayed puberty:** Kallmann syndrome.

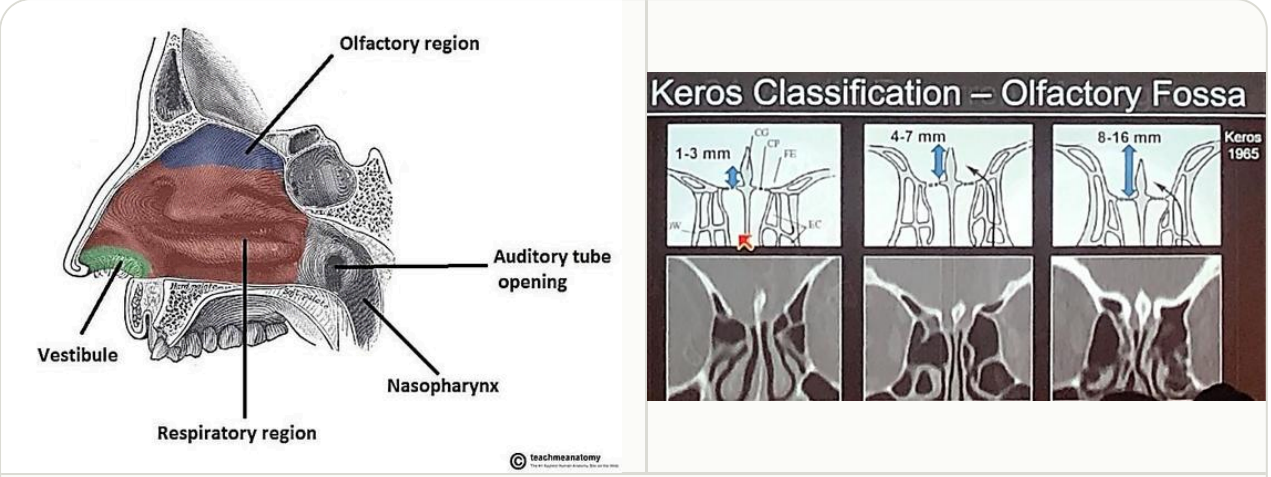


FIGURE 1 First, the olfactory region of the nasal cavity. Second, the Keros classification of the olfactory fossa.

Anatomy atlas · ENT Steps p. 97, ENT Steps p. 143

2 CHAPTER TWO History and examination

Pattern and cause.

2.1 History

Ask whether the loss came on suddenly or gradually, whether it fluctuates, and what preceded it: a cold or COVID-19, a head injury, nasal surgery. **Fluctuating** loss

that improves with steroids is conductive and inflammatory, typically chronic rhinosinusitis with polyps. **Sudden** loss after a viral infection or a blow to the head is sensorineural. Ask about distortions and phantom smells, nasal symptoms, smoking, chemical exposure, drugs, neurological symptoms and family history, and about safety at home (smoke alarms, gas detectors, food dates).

2.2 Examination and testing

Examine the nose with an endoscope, paying particular attention to the olfactory clefts, which may be blocked by polyps or oedema when the rest of the nose looks normal. Examine the cranial nerves.

Smell should be measured, not estimated. Validated tests include the 40-item **University of Pennsylvania Smell Identification Test (UPSIT)**, the **Sniffin' Sticks** (threshold, discrimination and identification, giving a TDI score) and short screening tests. Smell tests also detect malingering, because a patient who scores well below chance must be recognising the odours to avoid them.

2.3 Imaging

MRI of the anterior skull base and brain is indicated for unexplained loss, unilateral loss, or neurological symptoms; it may show reduced olfactory bulb volume. CT of the sinuses is indicated when chronic rhinosinusitis or a sinonasal mass is suspected.

3 CHAPTER THREE Causes

Conductive and sensorineural.

CAUSES OF SMELL LOSS

TYPE	CAUSE	FEATURES
Conductive	Chronic rhinosinusitis with polyps	Fluctuating, improves with steroids; commonest cause in clinic

TYPE	CAUSE	FEATURES
Conductive	Allergic rhinitis, septal deviation, tumours	Blockage; unilateral if a mass
Sensorineural	Post-viral, including COVID-19	Sudden after infection; parosmia common
Sensorineural	Head injury	Shearing of olfactory fibres at the cribriform plate; worse with occipital blows
Sensorineural	Neurodegeneration	Parkinson's (about 90% at diagnosis), Alzheimer's
Sensorineural	Ageing (presbyosmia)	Common over 65
Sensorineural	Toxins and drugs	Smoking, chemicals, radiotherapy, some drugs
Sensorineural	Congenital	Kallmann syndrome, isolated congenital anosmia
Other	Idiopathic	About a fifth of specialist clinic patients

3.1 Post-viral and COVID-19 smell loss

Viral damage to the supporting cells and receptor neurons is the commonest sensorineural cause. COVID-19 typically caused a sudden, complete anosmia that patients noticed sharply, and most recovered within weeks, but a substantial minority were left with lasting loss and parosmia.

3.2 Post-traumatic smell loss

Head injury shears the olfactory fibres at the cribriform plate or bruises the olfactory bulbs and frontal lobes. Recovery is less likely than after viral loss; only about a third to a half notice improvement over years.

4 CHAPTER FOUR Management

Treat the cause; train the nose.

4.1 Conductive loss

Chronic rhinosinusitis with polyps is treated with saline and topical steroids, preferably as drops in the head-down position, with a short course of oral steroids that also shows whether the loss is reversible. Endoscopic sinus surgery opens the olfactory clefts, and biologics (such as dupilumab) can restore smell in severe polyp disease.

4.2 Sensorineural loss

Olfactory training is the best-supported treatment: the patient sniffs four strong odours (traditionally rose, lemon, eucalyptus and clove) for about 20 seconds each, twice a day, for at least three to six months. Changing the odours every few months may improve results. Oral steroids are most likely to help early after a viral or traumatic loss, and topical steroids when the olfactory cleft is swollen. Other treatments, including omega-3 supplements, vitamin A drops and platelet-rich plasma, have limited evidence.

Every patient needs safety advice: smoke and gas detectors, checking food dates, and care with personal hygiene. Recovery after a viral loss can take years, and parosmia often signals regeneration.

SMELL LOSS

1 Red flags?

→ **unilateral with blockage or bleeding**
endoscopy and imaging, biopsy

→ **neurological symptoms**
MRI and neurology

2 Endoscopy of the olfactory clefts

→ **polyps or oedema**
conductive — steroids, chronic rhinosinusitis pathway, surgery or biologic

→ **normal**
sensorineural, step 3

3 Measure smell

→ **anosmia or hyposmia confirmed**
identify the cause — viral, trauma, ageing, neurological, congenital

→ **scores below chance**
consider non-organic loss

4 Sensorineural loss

→ olfactory training for at least three to six months

→ oral and topical steroids early

→ safety advice

→ MRI if no clear cause

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Conductive versus sensorineural loss;** the olfactory cleft.
- **Anosmia, hyposmia, parosmia and phantosmia.**
- **Validated smell tests:** UPSIT and Sniffin' Sticks.
- **Olfactory training:** the method and evidence.
- **Smell loss as an early sign** of Parkinson's disease.

- **Safety advice** for patients with anosmia.

5.2 Common pitfalls

COMMON PITFALLS

- Not examining the olfactory clefts.
- Telling a patient with post-viral loss that nothing can be done, when training helps.
- Missing a unilateral tumour behind "polyps".

Written for this app from Oxford Textbook of Otolaryngology ch. 26, Oxford Specialist Handbook ch. 3, Oxford Handbook of ENT ch. 3, Dhingra ch. 28, K. J. Lee ch. 30. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

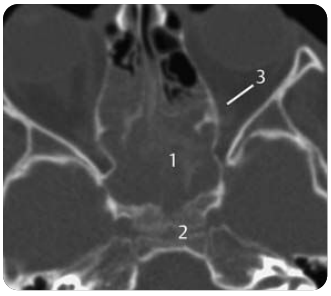
On imaging

The imaging of this topic: key findings, differentials and representative scans.

NOSE & NASAL CAVITY · TUMOURS

Olfactory Neuroblastoma (Olfactory neuroblastoma (esthesioneuroblastoma))

- Olfactory neuroblastoma (esthesioneuroblastoma) is an uncommon tumor of neural crest origin that arises from the olfactory mucosa in the superior nasal cavity. These polypoid tumors typically start unilaterally and may bleed profusely on biopsy. Neglected cases appear as a bilateral mass in the anterior cranial fossa. In these larger tumors, which usually have intracranial extension, peripheral tumoral cysts can occur at the margins of the intracranial part of the mass. These cysts have their broadest base on the tumor and, when present, are highly suggestive of the (imaging) diagnosis of olfactory neuroblastoma. Beware of encroachment of cranial nerves or intracranial vessels, which may be important in the estimation of morbidity during removal.



CT, axial bone window. There is complete bilateral opacification of the ethmoid and sphenoid sinuses (1). The anterior cortex of the clivus (2) seem…

NOSE & NASAL CAVITY · INFECTION & INFLAMMATION

Nasal Polyps

IMAGING FINDINGS

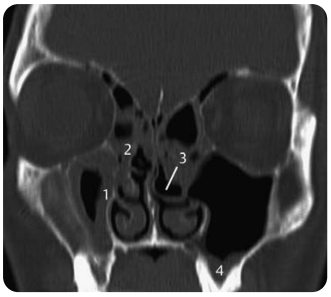
- Polypoid soft tissue filling the nose and all sinuses, both sides
- Widening of the ethmoid labyrinth and thinned septa
- Hyperdense contents suggest fungus or inspissated mucus

TOP DIFFERENTIALS

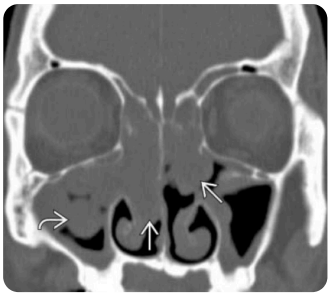
- Allergic fungal sinusitis
- Cystic fibrosis
- Inverted papilloma (unilateral)

CHECKLIST

- Bone erosion or asymmetry: think tumour or fungus
- Pre-operative CLOSE checklist



CT, coronal. The scan shows mucosal thickening in the right maxillary sinus (1), partial opacification of the ethmoid cells (2) on both sides, pneumatizati…



Coronal bone CT shows the classic appearance of sinonasal polyposis

Koch et al., Diagnostic Imaging: Head and Neck, 3rd ed., p. 708

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 708.